

Unit 1 Transcript

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Hi, everyone.

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Welcome to your very first step into the world of programming.

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My name is Demusomathokuti.

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I'll be one of your co-instructors in the course.

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I hold a degree in computer science and informatics.

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I'm currently pursuing my honours degree in computer science from the University of Johannesburg.

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I use Python in my honours to build artificial intelligence and computer vision systems.

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My message to everyone who's new here, I know you might be a little bit intimidated and don't know what to expect, but really I would like for you to remain calm and we will take you to the course step by step.

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so you will learn everything you need to know.

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And by the end of this short video, you will have your environment set up and ready to create magic.

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To make sure everyone's journey is smooth during the course, I'm going to help you

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to download Python and VS code into your machine so what I would like for you to go to do is to go to one of your browsers it can either be Firefox it can either be Microsoft Edge for the people with the Edge and or Google Chrome so currently now for demonstration I'm using Google Chrome so I want you to go to Google Chrome and search

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python right so the very first link you will see is python.org I want you to click on it so this is what you will see when you go to the website I want you to go then click on download so you'll see download the latest version for Windows for Windows people

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you don't click for download Python install manager I want you to go to get the standalone installer which is Python 3.14.6 which will be the latest version you will get and for people using different operating systems like your Linux, Mac OS

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Android, iOS and others you can just click on your operating system here then you'll be able to find the latest version of Python for Windows people we click so the moment you click you can see Python will start downloading let's wait for it for a second to finish with the download

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again once the executable file is downloaded click open okay so this is the first thing you'll see when you click the executable file one thing I would like for us to do this is very important add python executable to path if you don't do this you might

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I get in a bit of a trouble if you don't understand how to set your environment variables.

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That's where you will be able to set your path.

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But since we click, then just install.

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Then we will wait for it a bit to install.

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It will take a second.

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Okay, so setup was successful and we can close.

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So to verify if our Python was indeed installed, what you're going to do is going to go at the bottom left of your screen to search.

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So when you get to the search space, type CMD, right?

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After typing CMD, just click enter, right?

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This is what you will see.

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This is our terminal, command line terminal.

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So what I want you to do is to write Python, space, slash, slash, and say virgin.

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Right, then click enter.

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So what you're supposed to see is Python 3.14.6.

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or whatever the version of Python installed, right?

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If you type Python version and you don't see Python with some number here, that means you did the installation incorrectly.

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But for now, we can see Python was successfully downloaded.

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On to the next step.

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okay on our next step what are we going to what are we uh going to try to do is to download our code editor uh our code editor is where we are going to write our code and execute our code like run our code to see it give us some things like output so

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for our code editor I want us to go and download Visual Studio code so let's add here then let's search Visual Studio code this you go back to your web browser and search Visual Studio code say enter and the first link we get

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okay fine okay once you go to the link this is what you are seeing right if you use Windows you can just directly click here download for Windows but if you use the other operating system you can come to other platforms then you can see all the other platforms that are available Windows your Linux and Mac OS so since I'm using Windows

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I'm just gonna click here so once I've clicked there my file start downloading so I will be waiting for it to finish don't worry they might take a second to download

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Let's wait a bit.

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Almost done.

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Okay, so our file is downloaded.

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Let's go to our file.

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Once it's downloaded, you will read the license.

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Nobody ever does, which is not a good thing not to do.

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next then it will show you where the file will be stored I will advise that you don't change this just take the path that they give you here or you can browse and change the path if you want to just click next Visual Studio Code next additional icons create a desktop icons like next

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Then install.

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Installation will take a second.

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Let's wait for it.

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Our installation is complete.

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We click finish.

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So moment you click finish, it will open your Visual Studio Code.

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And yes, this is what you see when you open VS Code.

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I know it might be a little bit intimidating to look at, but

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should not be worried about anything we will go through this explaining what each tab does and what to do for you what you need to do for you to be successful in using Visual Studio Code so the first thing I would like for us to do is to change the theme it's a little bit dark let's go and try work with something

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visible again so you go down the left bottom corner for the settings right so um after clicking uh the the manage button I want to go to go to themes uh color theme then let's say light 2026 yeah now things are brighter and we are moving forward

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so once we've changed the theme the first thing I also want us to do is to go to the left the top left and click on the bottom icon with the blocks and the other one shifting this is where we get our extensions right so what our what is or what are extensions

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extensions are the software that help us achieve things they can either help with the wallpaper the way things look or the way VS code interact with certain languages since we are going to utilize Python we have to go and download a Python extension the Microsoft one you can see this let's click on it

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Then install, so this Python will give VS Code the brains to understand the Python code in terms of the keywords, in terms of...

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uh the different functions we have so that you can represent them using different colors and be able to understand the Python language so uh get started with Python mark is done okay so our extension is downloaded so now I would like for us to go through

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Visual Studio Code okay ladies and gentlemen let us go through Visual Studio Code editor so starting from the top like you see a file at the top so when you click a file you're given a few functionality uh

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different functions you can work with this is whereby you can create your new files you can open folders that you already created you can add other folders to different workspaces duplicate workspaces and you can close the windows and for editing editing is used on the current working file

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You can do the undo, redo after writing to a file.

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You can cut, copy and paste.

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You can write the blog comments using the edit function.

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For selection, there's also you use selection on the current open file, like which part of the file you would like to select, adding a cursor

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on certain lines and so forth view view is uh what type of uh what do you call window you like open the different types of view you can have an explorer um search right so these are all the things that you see this side your extension do you want to view um the testing the last one

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this is you can access them to view you can also add and remove them from uh the left hand side so go we use go to go to different places it's either in a file so like different lines you want to go to specific line right you want to go to a definition

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inside a file, then run, this is what we use to execute our files, like run this script, then it will be able to give you an output.

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The output is released on what we call the terminal, right?

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So you can, we can create a terminal.

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for you to be able to see.

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So at the bottom, you just see this part just popped up.

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This is what we call a terminal.

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Terminal usually just give you a file path where we are in.

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And also it outputs results here.

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So all your output, you will see them in a terminal.

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You can be able to change the specific type of terminal you want.

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You can create a new terminal again.

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You can split the terminals.

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You can select what type of terminal you want.

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Is it PowerShell?

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Is it command prompt?

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Is it JavaScript debug terminal?

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Yeah, you are spoiled for choice.

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So, and also this the last one.

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help is if you don't understand some of the things here, you are always allowed to go to help in order to get a clear understanding of what VS Code is about.

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You can also join the VS Code.

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They have their own YouTube channel where they basically take you through all the functions that they have.

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I think it's advisable for you to go and look for yourself.

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to get more clarification on certain things, right?

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Then we can go on the farther left.

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You can just click here.

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The first one is in Explorer.

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For Explorer, this is where you see the current open folder or a file.

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Then you can always search if you are looking for things.

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Then we have source control.

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Source control is how

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We manage our files, right?

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And also source control gives you a git.

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You're able to use git in order to upload, pull and push code.

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It also allows you to use GitHub, working with multiple coders.

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Then we have the run and debug function.

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which is self-explanatory use this when you run and at the same time want to debug the code you just ran check for your mistakes and bugs you might find in the code then the extension we already went here you can get the different types of extension from here not just Python related things can be any other different languages

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It can be some software to change how VS Code looks.

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There are a bunch of things.

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Then we have a test now where you can test your script.

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All right.

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So closing this.

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Okay.

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So don't know if you've seen we have a welcome page.

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that you get as soon as you open VS Code, instead of you having to go to file and creating a new file, open a file, creating a new file, like here instance, you can get an option to create a new file here on the work home page, opening a file, opening a folder, right?

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There are a bunch of functionalities you find here.

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and on the further right you have a chat box which is an AI tool used to help you how to code but as of now this is currently bad for you and I would advise for you to go at this corner and just click here whenever you see this right yep so I think this is it with

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VS Code.

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One thing I would like for us to do is to actually create something using VS Code, right?

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So one thing I would like for us to do is to create

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okay a new folder then we can do this open folder okay so you can just say open a folder I would like for us to go to desktop uh can right click here so it will show you this menu you create a new folder named uh

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VS code.

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Let's just name our folder VS code.

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Then we can select the folder.

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Select the folder.

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Yeah.

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As you can see, the moment you create the folder, then the explorer is open.

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As I've explained previously, this is where you will see your folders and all the files you create, right?

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Then the moment the VS Code folder is created, you are given an option.

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The first option you have is to create a new file inside that VS Code folder or create another folder inside that folder.

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But what we would like to do is to create a new file inside that folder.

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So our folder name

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Our file name will be, let's just name it python.py.

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It's not a must to name it a python, you can name it anything.

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But it is very important for you when you name your file to have that dot py.

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right to make uh vs code understand that this is a python file we are working with a python file here then after writing the name just press enter right can close this we can click here to close this right so after we've created our first uh python file we can try and

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print something like hello well this is the first line of code we wrote together right so we can just say run dust and continue uh then you will select the python uh debugger right

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So after selecting your Python debugger, you will see the VS code will avail this play button, which is a button you are going to use to run Python script, right?

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Since we printed hello world, we are expected to do what?

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To see hello world as an output, right?

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So it already ran this.

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because at the top there I ran the code so this is what you see hello world as an output we can change this just for demonstration hello world okay then exclamation mark then let's save our code then we run again let's run okay

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as you can see down here our output is hello world again and I think that will be it for the first video then expect more on the next video where we start getting to learn Python

as a language just going through the basics of Python and we'll pick it up from there
thank you